MICROSOFT
Data Analyst

Role Practice Assessment

20 original prompts with space to plan your responses

Scenario lens: Fictional product work related to cloud and collaboration. This pack uses general skills expected in data analyst work. It is independently written, is not an official hiring test, and is not affiliated with Microsoft.

20 prompts	75-90 minutes	100-point review guide
Five skill areas	Show your reasoning	Use the rubric at the end

How to use this pack

Answer each prompt in the space provided. State assumptions, show calculations or pseudocode where useful, and explain trade-offs. Use a separate sheet if you need more room. The focus labels are a self-review guide, not a company scoring key.

Suggested review rubric

Technical correctness 35 points · reasoning and trade-offs 25 · validation and edge cases 20 · clarity and communication 20

Name: _____ Date: _____

SQL and data modeling

Fictional cloud and collaboration scenario · answer all four prompts in this section.

01

From an event table (user_id, event_name, event_time, session_id), calculate daily active users for cloud and collaboration. Define deduplication and timezone rules.

Focus: Correct grain, distinct users, date boundaries

02

Write a cohort-retention query using a user's first activity month and monthly return activity. Explain how you handle incomplete months.

Focus: Cohorts, date arithmetic, censoring

03

A join doubles revenue because one order has several item rows and several status events. Diagnose and repair the query grain.

Focus: Join cardinality, pre-aggregation, fact grain

04

Find the top three categories by weekly revenue, including ties. Describe a query using window functions.

Focus: Aggregation, ranking, tie behavior

Data quality and instrumentation

Fictional cloud and collaboration scenario · answer all four prompts in this section.

05 A dashboard's cloud and collaboration events suddenly fall by 30%, but backend transactions are stable. List checks to find the break in the data pipeline.

Focus: Freshness, schema, volume, client/server comparison

06 Design a data-quality checklist for a new checkout or subscription event. Include required fields and validation rules.

Focus: Event contract, nullability, allowed values, ownership

07 Two systems report different order totals. Plan a reconciliation that finds the source of the mismatch.

Focus: Definitions, grain, time windows, traceable samples

08 A numeric field contains missing values, zeros, and extreme values. Explain how you investigate before choosing a treatment.

Focus: Missingness, validity, robust summaries, no blind imputation

Metrics and experimentation

Fictional cloud and collaboration scenario · answer all four prompts in this section.

09 Define a funnel for a user discovering and completing a core task in cloud and collaboration. State the numerator and denominator at every step.

Focus: Mutually clear steps, eligible population, conversion

10 Plan an experiment for a redesigned results page. Specify randomization unit, primary metric, guardrails, and exposure rules.

Focus: Randomization, metric choice, guardrails, contamination

11 An experiment shows a small positive lift with a wide confidence interval. Explain what you would tell a product lead and what you would do next.

Focus: Uncertainty, practical significance, decision framing

12 Check an experiment for sample-ratio mismatch and novelty effects. Describe the diagnostics and possible remedies.

Focus: Allocation check, time trends, instrumentation

Product and business analysis

Fictional cloud and collaboration scenario · answer all four prompts in this section.

13 Weekly engagement is down for cloud and collaboration. Break the change into plausible segments and prioritize the first three analyses.

Focus: Decomposition, cohort/segment reasoning, prioritization

14 Choose a north-star metric and two guardrails for a feature that helps users find relevant content or products.

Focus: Value, quality, safety or cost trade-offs

15 Revenue rose while the number of paying users fell. Give several hypotheses and the data needed to distinguish them.

Focus: Mix, pricing, retention, acquisition, definitions

16 Compare two markets with different user mixes. Explain how Simpson's paradox could reverse the overall result.

Focus: Stratification, weighting, comparable populations

Communication and decision quality

Fictional cloud and collaboration scenario · answer all four prompts in this section.

17 Turn a messy analysis into a one-page recommendation. Specify the headline, evidence, caveat, and proposed decision.

Focus: Clear narrative, traceability, caveats, action

18 A stakeholder asks for a single success number before the metric is defined. What questions do you ask first?

Focus: Decision context, population, timeframe, metric contract

19 Create a dashboard outline for a product manager. Choose a small set of leading and lagging indicators and explain each.

Focus: Audience, hierarchy, trends, actionable thresholds

20 Explain a statistically significant but tiny result to a non-technical audience. Recommend a responsible next step.

Focus: Plain language, effect size, uncertainty, practical value

Self-review and submission checklist

Before sharing your work, use this independent review guide. It is not an official company rubric.

Area	Points	Check your response
Technical correctness	35	Does the solution answer the prompt and use sound methods?
Reasoning and trade-offs	25	Are assumptions and alternatives explained?
Validation and edge cases	20	Did you check data quality, boundaries, and failure cases?
Clarity and communication	20	Can a teammate follow the result and proposed next step?
Total	**100**	Use feedback to choose what to practice next.

My strongest area

One skill I will practice next